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Leveraging LLMs and a Common Data Model to Unlock Real World Evidence from Unstructured Electronic Health Record Data in India

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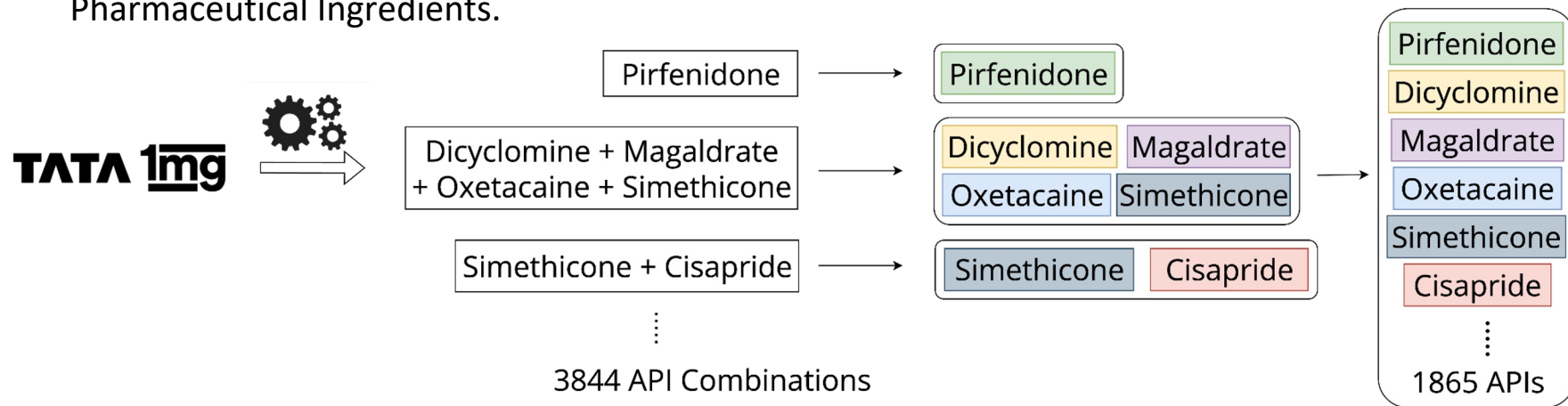
Aim

- To analyse current antibiotic prescription patterns in India among outpatients with febrile respiratory illness
- To leverage open source Large Language Models and NLP algorithms to extract, transform and load clinical features from unstructured clinical notes into the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)

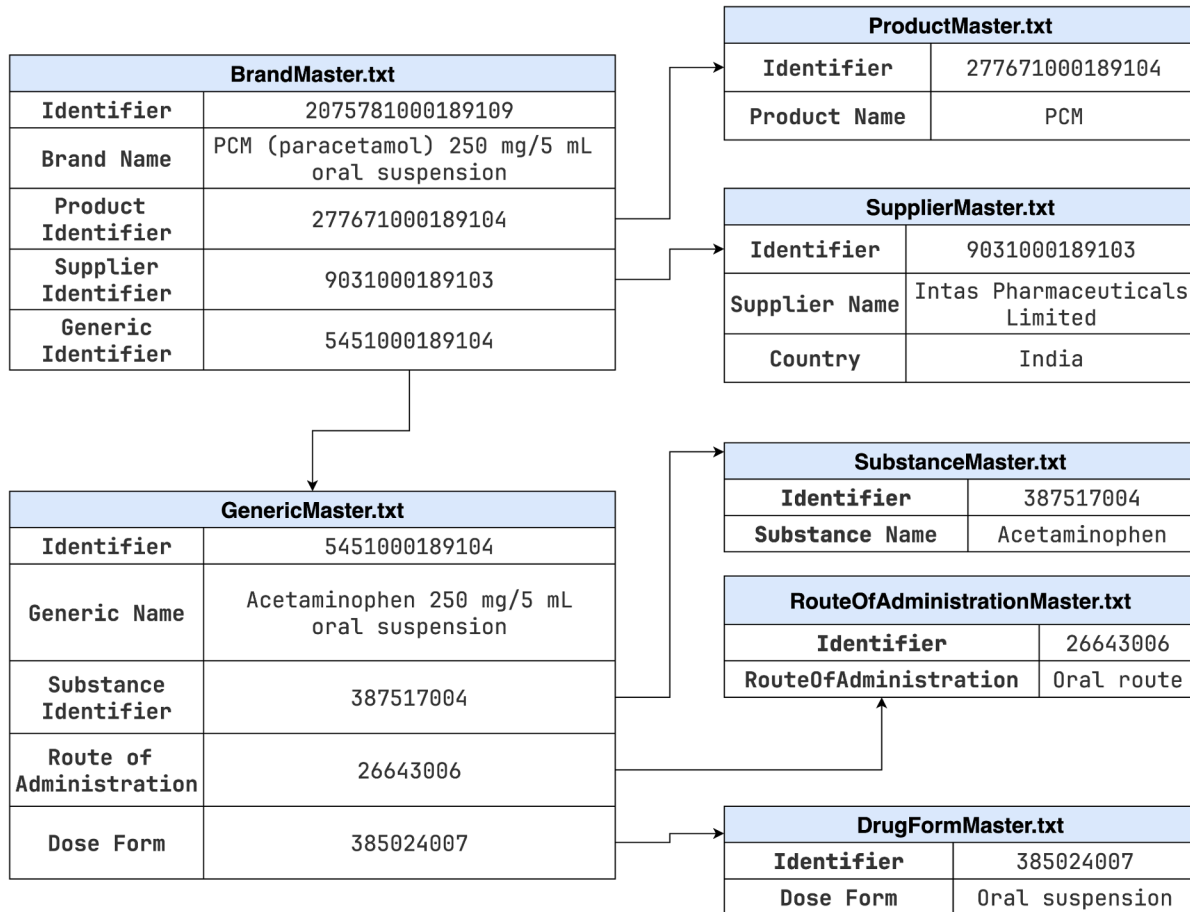
- Intended analysis is difficult as current open-source Large Language Models **perform poorly** on tasks involving medication terms specific to Indian settings.
- A comprehensive, standardised **knowledge base** of Indian medication terms is required that can be leveraged by Language Models (through pre-training or fine-tuning or in-context learning) for better performance on clinical notes in Indian settings.

Knowledge base development

- To develop a comprehensive, interoperable knowledge base, we first need a list of all medication terms specific to India.
- Amongst the many ecommerce platforms in India selling both prescription-based and over-the-counter medication, we extracted medication terms from 1mg.com.
- The extraction process resulted in lists of **3,63,525** Brand Drug names and **1865** unique Active Pharmaceutical Ingredients.

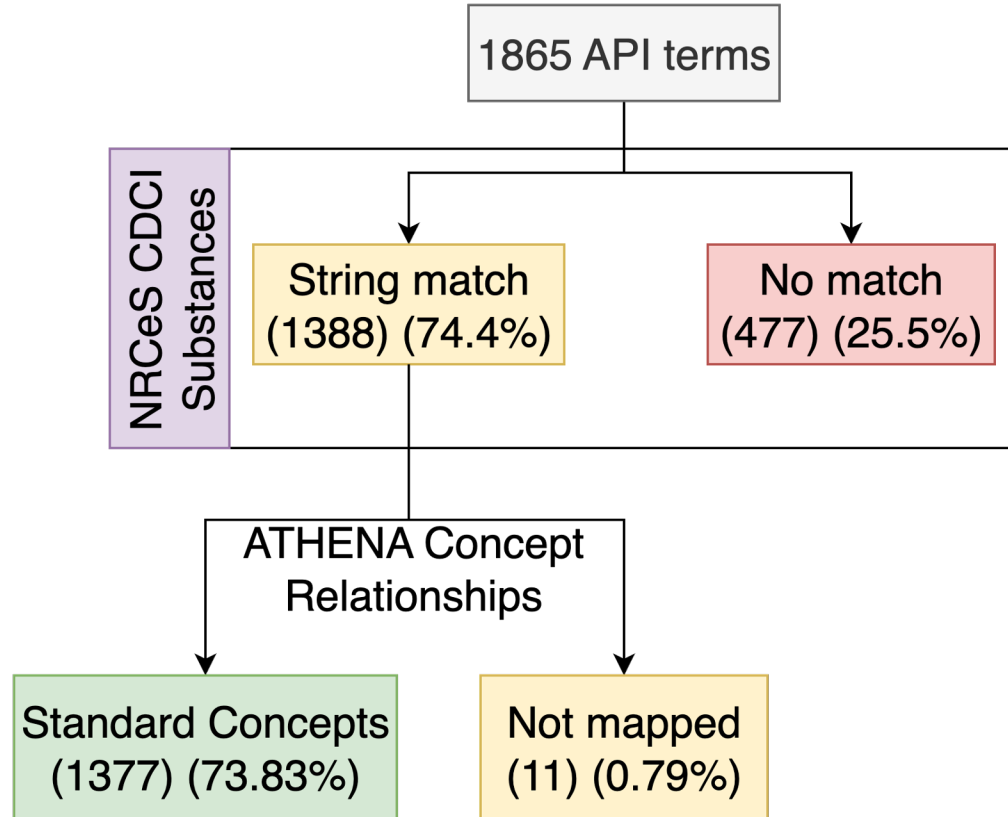


NRCeS Common Drug Codes for India tables



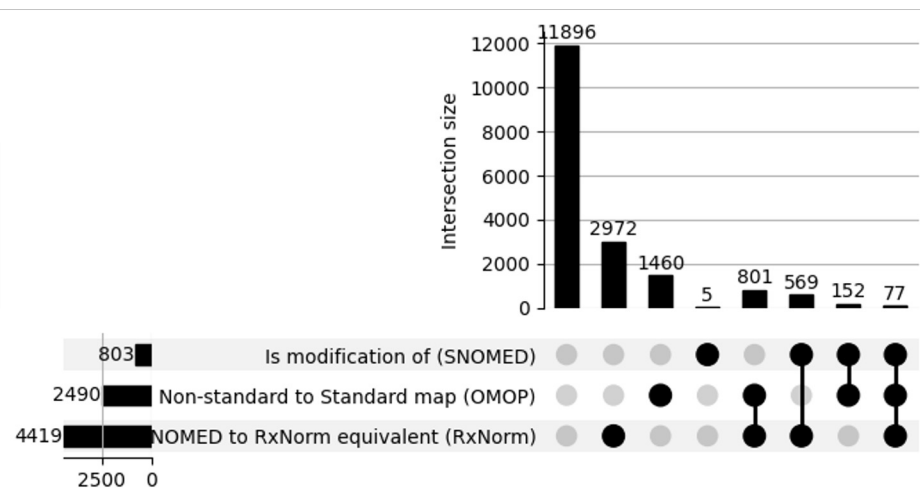
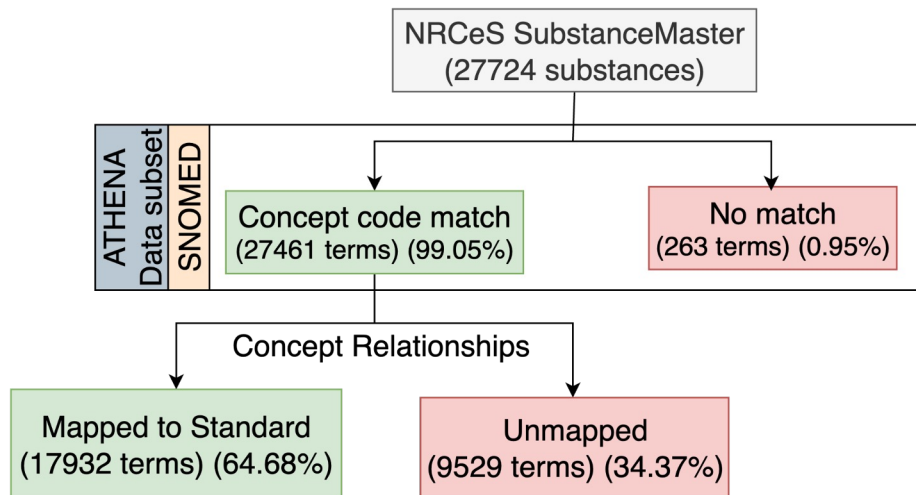
Knowledge base development

- We map the API terms to the NRCeS CDCI Substances table by string matching.
- Next, we see how many of these terms (mapped to NRCeS CDCI) were mapped to standard.



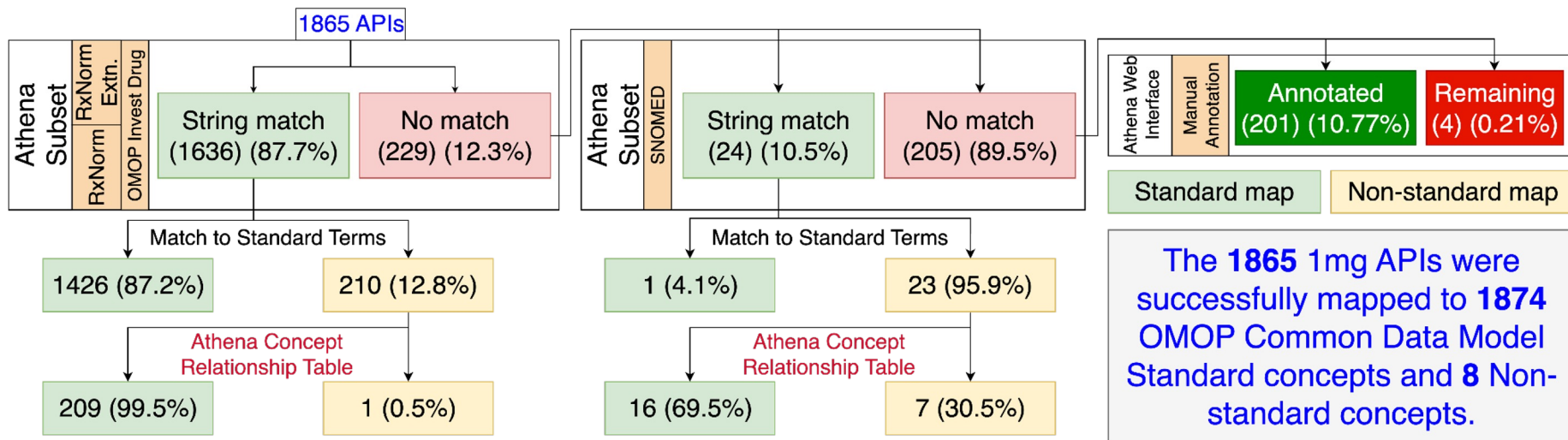
Knowledge base development

- We assess how many Substances in the NRCeS Substance table map to Standard using OHDSI Standardised Vocabularies and ATHENA Concept Relationships table

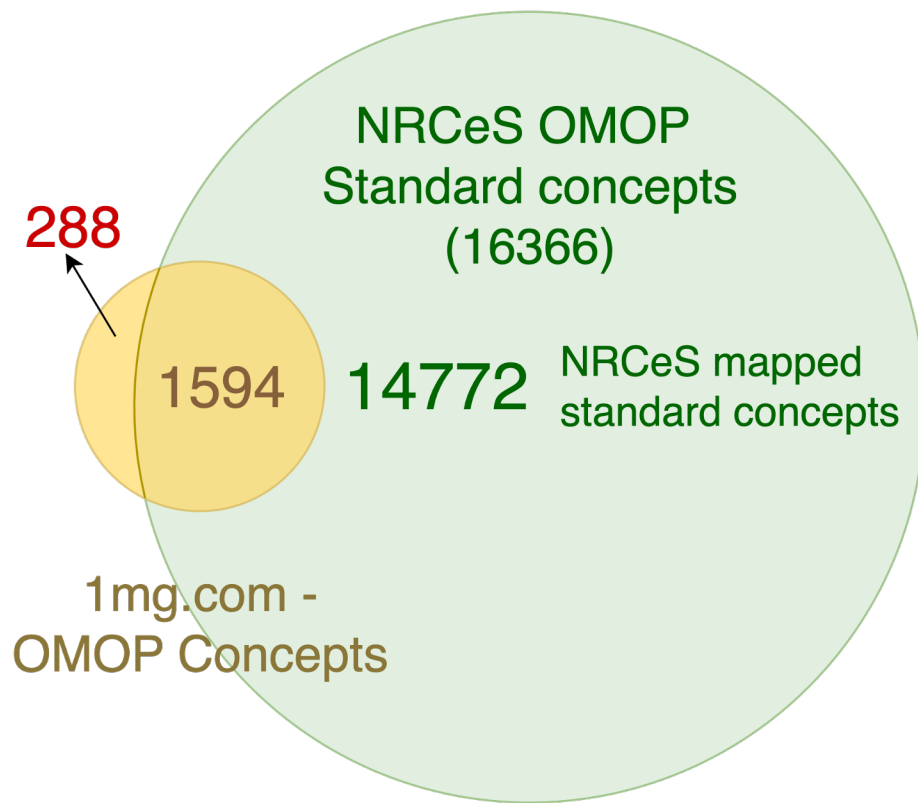


Knowledge base development

- Extracted API terms were mapped to OMOP Concepts using the OHDSI ATHENA data dump (August 2024) and the [ATHENA web interface](#) for manual annotation.



Intersection of NRCeS OMOP and 1mg.com OMOP concepts



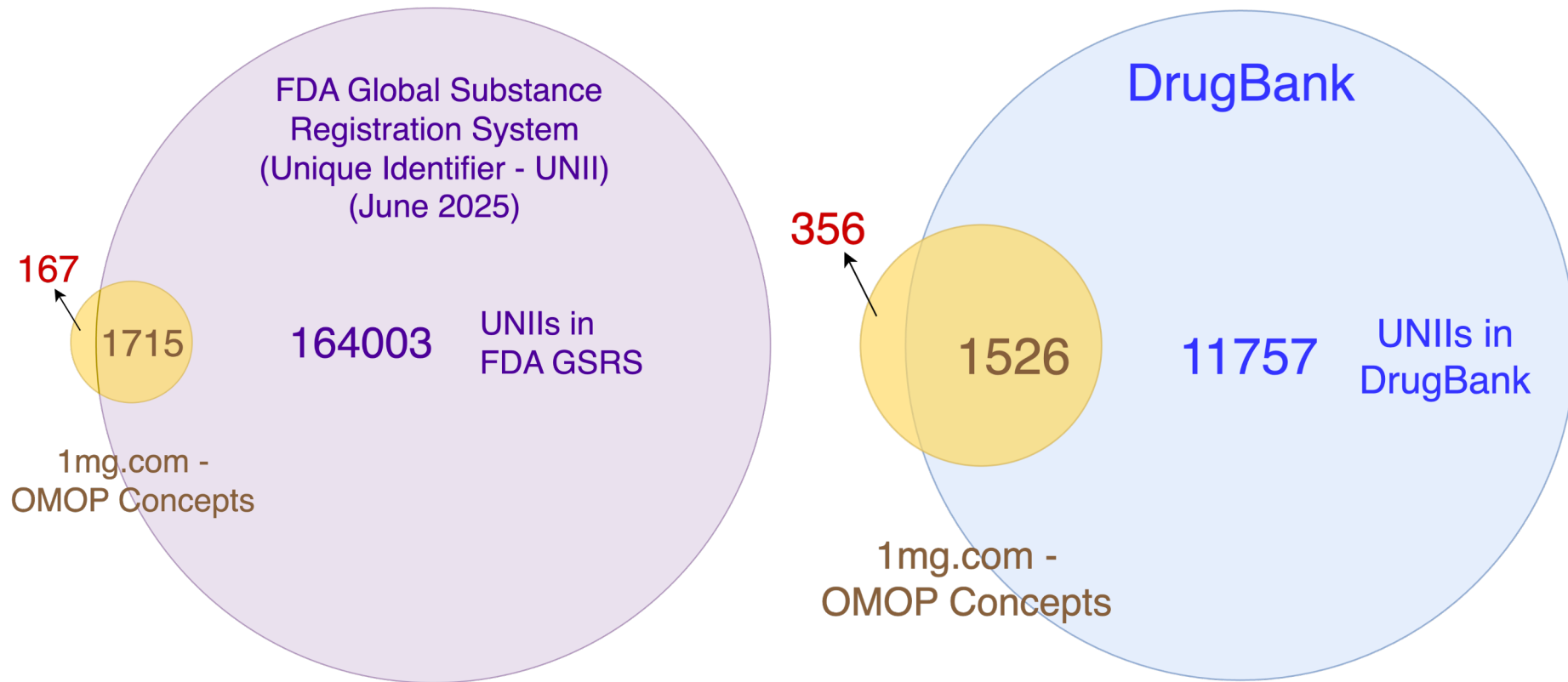
This venn diagram showcases intersection between mapped OMOP Concepts from NRCeS CDCI Substances and extracted API terms.

Linking to Global Databases

- **FDA's Global Substance Registration System (GSRS)** enables accurate exchange of information on substances through their Unique Ingredient Identifiers (UNII)
- The **UNII GSRS Records** file consists of unique identifiers from different Chemical, Biomedical and Taxonomic databases.
- The following figure showcases the different Identifiers collated in the **UNII GSRS Records** file.

General Identifiers	Biomedical Identifiers
Display Name	Unique Ingredient Identifier (UNII)
INGREDIENT TYPE	NCI Thesaurus Concept Code
	RxNorm Concept Code
	European Medicines Substance ID
	WHO-INN Identifier
	United States Adopted Names ID
	SUBSTANCE TYPE
	DAILYMED UNIIs
	FDA-SRS record UUID
Registry Number	Catalogue of Life organism ID
EPA Substance ID	ITIS Serial Number
PUBCHEM Compound ID	NCBI taxonomy organism ID
Molecular Formula	POWO Online
InChI Key	GRIN nomen ID
SMILES	MPNS Record ID
ECHS registry number	USDA PLANTS organism ID
Chemical Identifiers	Biological Taxonomy IDs

Linking to Global Databases



WHO AWaRe classification of antibiotics

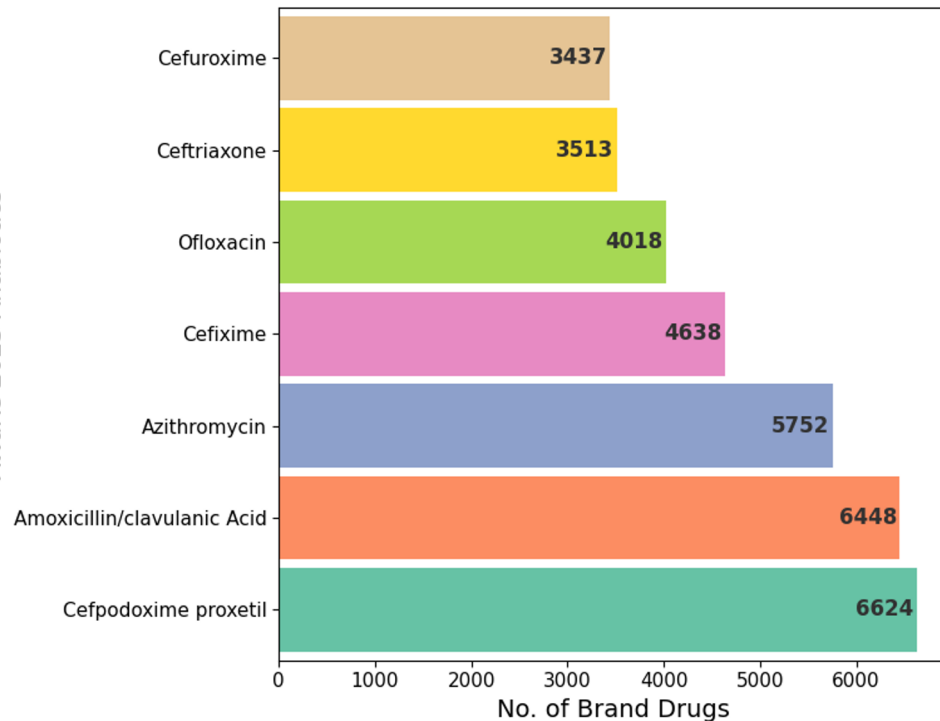
- The **AWaRe (Access, Watch, Reserve) classification of antibiotics** was developed in 2017 by the WHO Expert Committee on Selection and Use of Essential Medicines as a tool to support antibiotic stewardship efforts
- The classification of antibiotics is based on their susceptibility to inducing antimicrobial resistance, with Access antibiotics being the least susceptible and Reserve antibiotics being the most susceptible.
- The 2025 AWaRe antibiotics list was mapped to OMOP Standard Concepts, and then these were mapped directly to the 1mg.com extracted brand drugs.

Access	Watch	Reserve
92	146	30

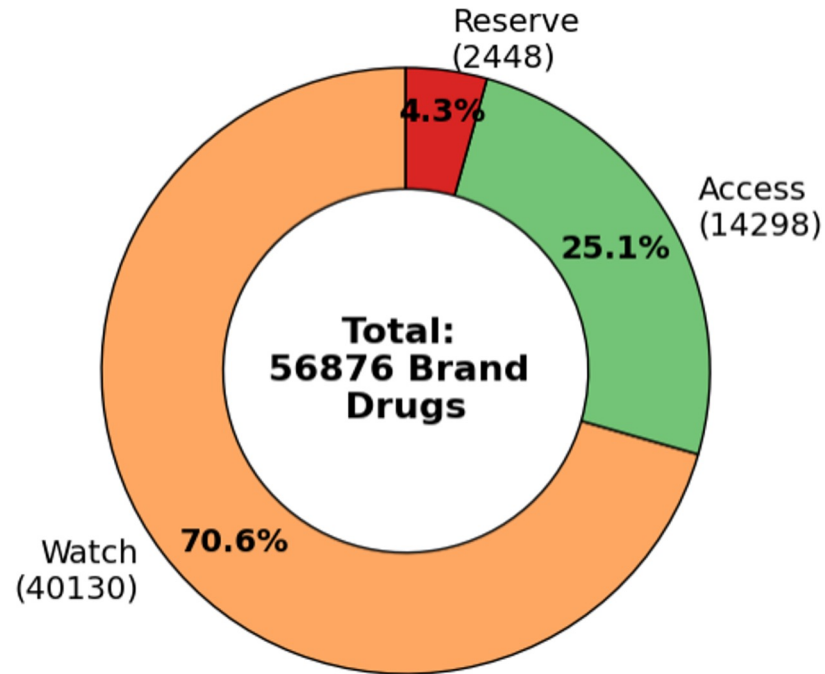
Table: No. of antibiotics for each AWaRe category

WHO AWaRe classification of antibiotics

AWaRe 2025 Antibiotics



(i) Top 7 Antibiotics based on number of brand drugs

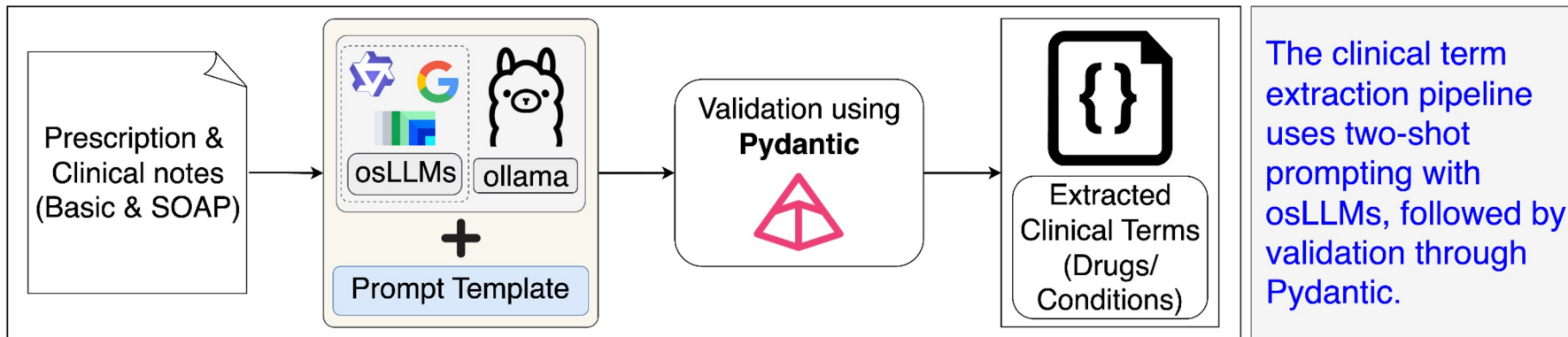


(ii) AWaRe antibiotics brand drugs segregated by AWaRe category

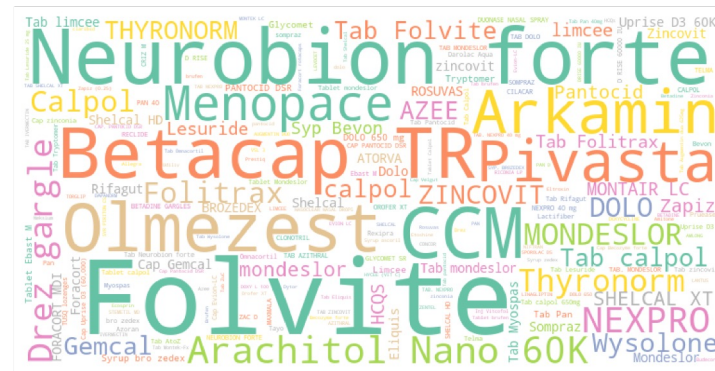
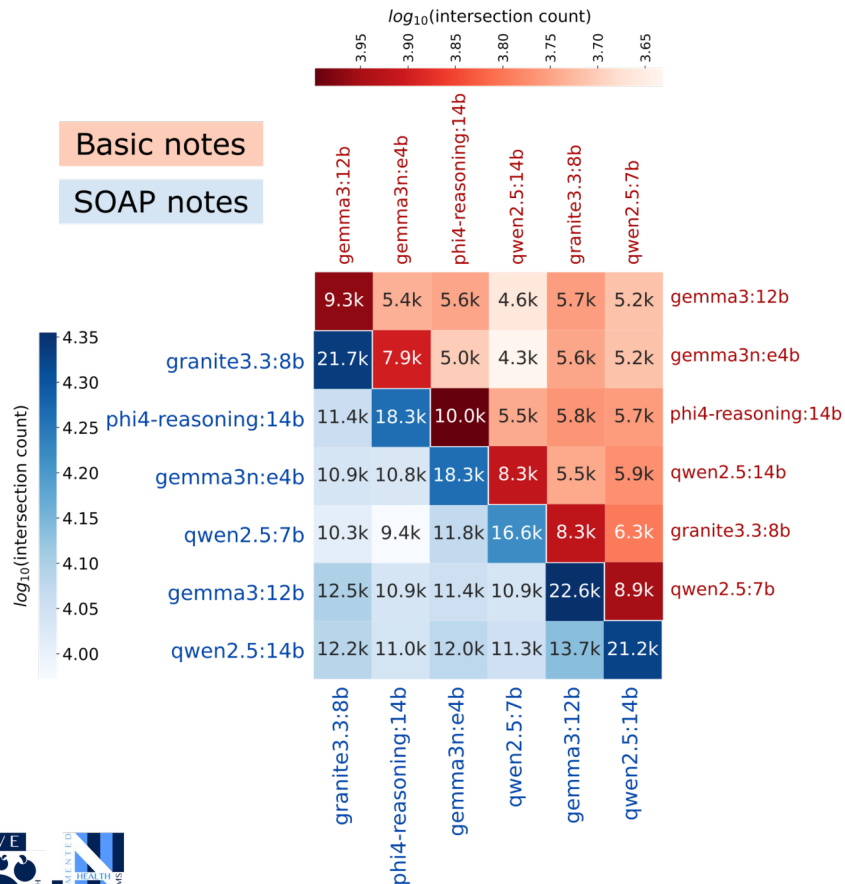
Clinical term extraction from unstructured clinical notes

- The dataset consisted of a total of **263774** unique records.

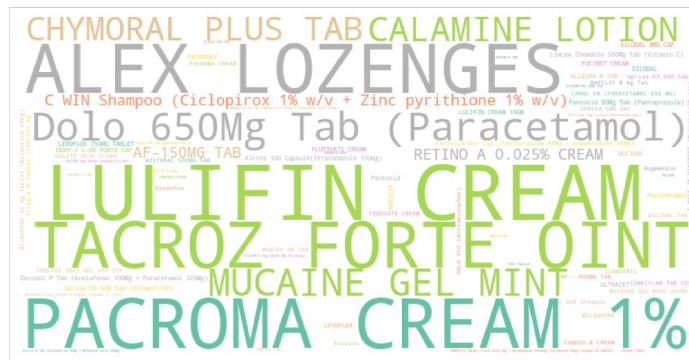
Prescription notes		Diagnosis notes		
Basic	SOAP	Basic	SOAP assessment	SOAP assessment notes
45723	26941	60972	16388	13002



Clinical term extraction from unstructured prescription notes



Basic notes extracted terms



SOAP notes extracted terms

Benchmarking Open source Language models for medication extraction task

Model	F1 Score	Hallucinated	Error Rate
Mistral v0.3 7B	0.81 ± 0.29	0.88 ± 1.27	0.15 ± 0.29
Gemma 2 9B	0.84 ± 0.31	0.38 ± 0.76	0.16 ± 0.31
Llama 3.1 8B	0.81 ± 0.33	0.49 ± 0.95	0.19 ± 0.33
Qwen 2 7B	0.77 ± 0.32	0.89 ± 1.21	0.20 ± 0.32
DeepSeek-R1 7B	0.76 ± 0.35	0.75 ± 1.14	0.24 ± 0.35

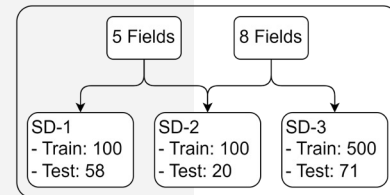
(i) Performance using prompt engineering.

(ii) Performance using parameter efficient fine-tuning.

Model	F1 Score	Hallucinated	Error Rate
Mistral v0.3 7B	0.82 ± 0.35	0.28 ± 0.57	0.18 ± 0.36
Gemma 2 9B	0.90 ± 0.23	0.21 ± 0.44	0.10 ± 0.24
Llama 3.1 8B	0.95 ± 0.18	0.13 ± 0.38	0.06 ± 0.20
Meditron 7B	0.93 ± 0.19	0.21 ± 0.63	0.06 ± 0.19
Llama 3 8B	0.92 ± 0.22	0.18 ± 0.52	0.08 ± 0.22
Mistral v0.1 7B	0.92 ± 0.22	0.27 ± 0.89	0.08 ± 0.22
Medalpaca 7B	0.80 ± 0.36	0.72 ± 1.70	0.21 ± 0.36
Llama 2 7B	0.65 ± 0.46	0.45 ± 0.56	0.35 ± 0.46

We have developed a high-quality annotated dataset of **800+ clinical prescription notes**, categorized based on different annotation schemes and timelines. The annotations contains following fields:

- 5 Fields:
 - **name, dosage_form, duration, frequency, strength**
- 8 Fields:
 - **drug_name, dose, dose_amount, dose_change, frequency, intake_time, route, duration**



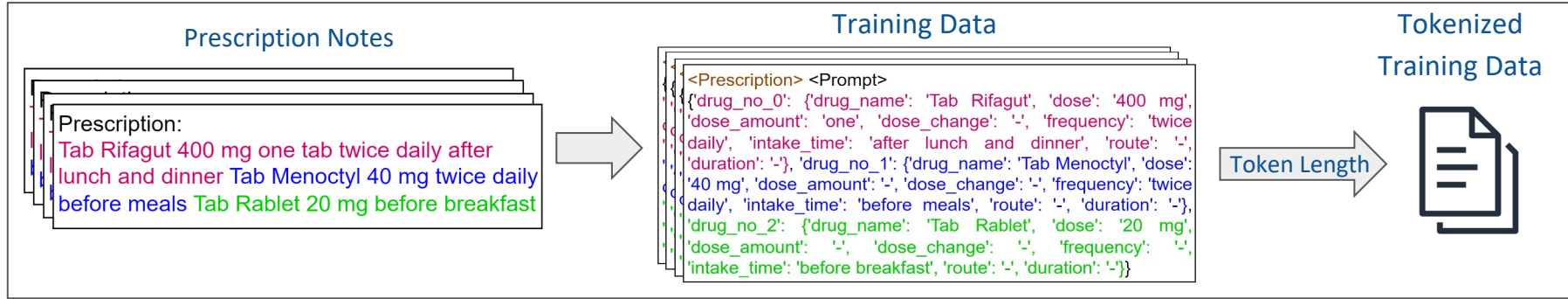
The dataset is structured into three sub-datasets (SD):

- SD-1: 158 notes (5 Fields)
 - SD-2: 120 notes (5 Fields + 8 Fields)
 - SD-3: 571 notes (8 Fields)
- Total Notes: **158 + (120 + 571) = 849**

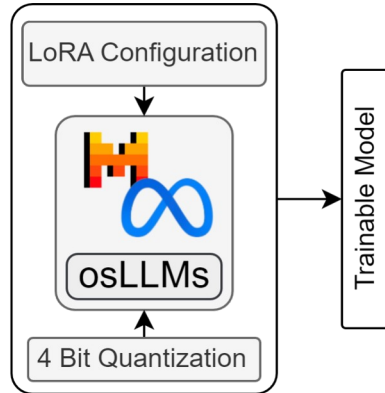
Key Distinction:

The primary difference between SD-1 and the combined SD-2/SD-3 datasets lies in the time and **type of prescription notes**. SD-1 was annotated earlier with a focus on the 5 fields and contains , whereas SD-2 and SD-3 were annotated later with 8 Fields.

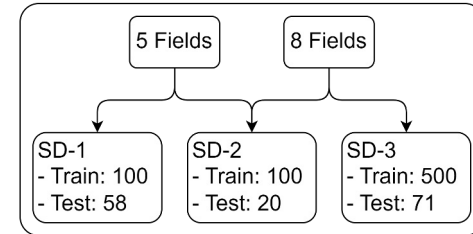
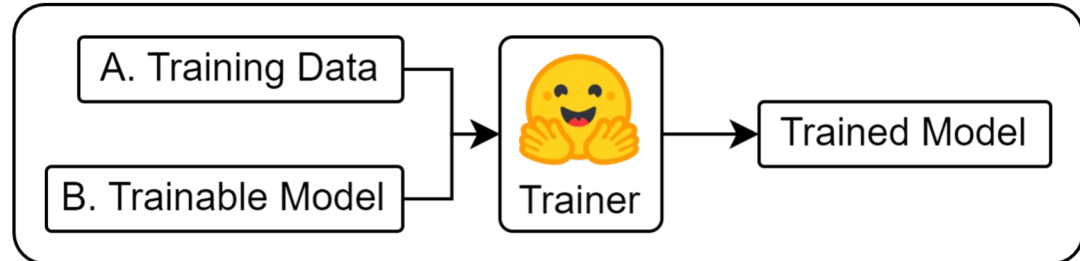
A. Training Data Preparation (500 training data and 71 test data)



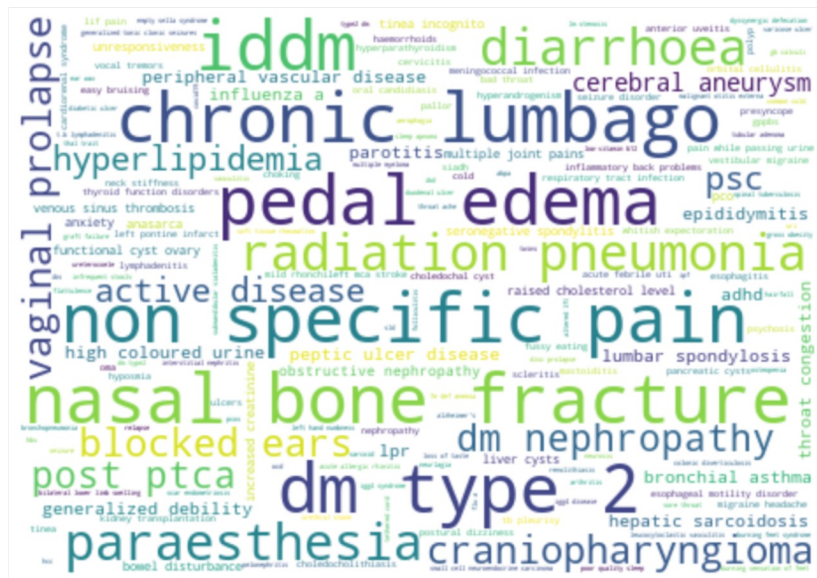
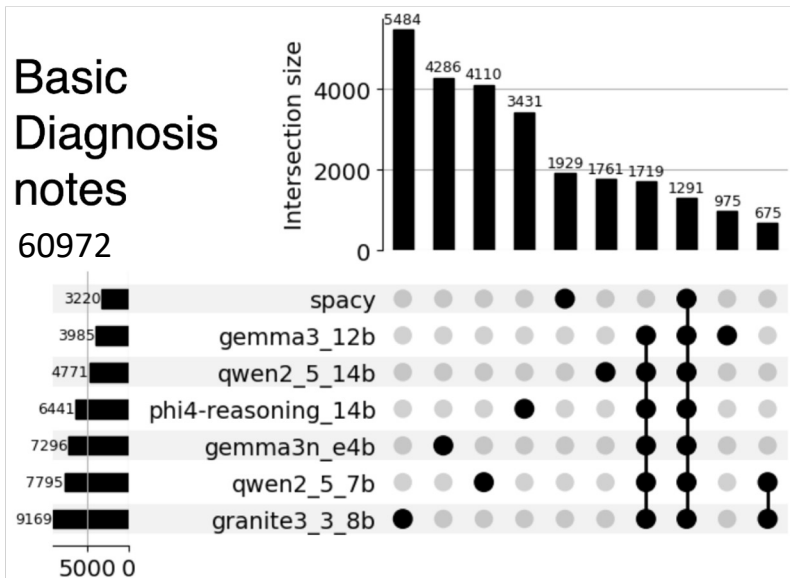
B. Model Preparation



C. Training



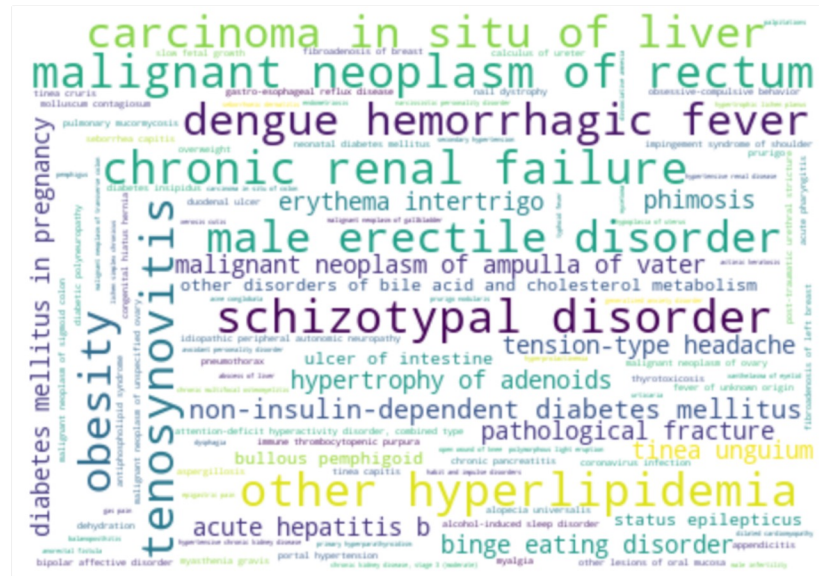
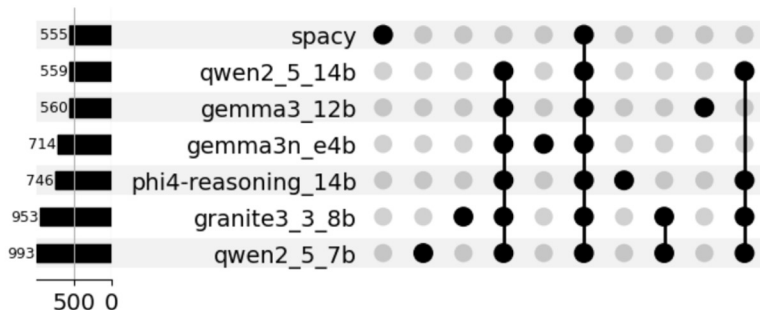
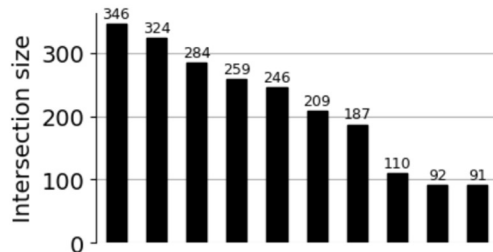
Clinical term extraction from unstructured diagnosis notes



Clinical term extraction from unstructured diagnosis notes

SOAP Assessment

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Clinical term extraction from unstructured diagnosis notes

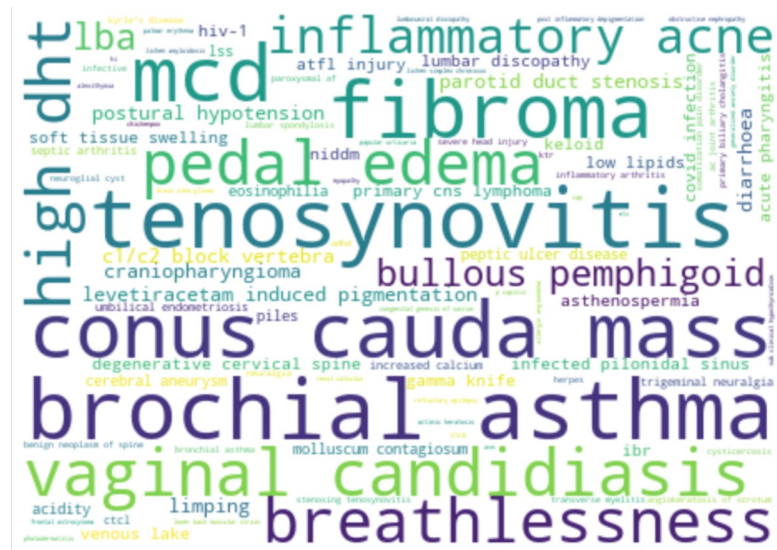
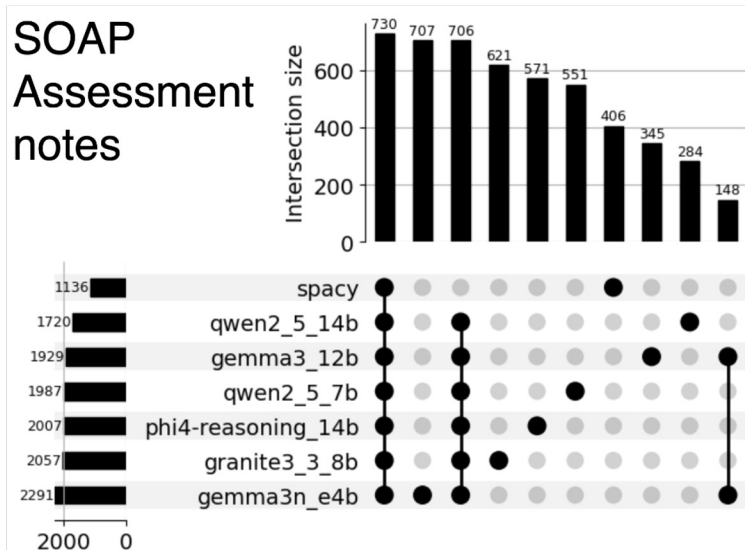


Table 2: Comparison between LLM and spaCy NER Models — Basic Notes

LLM Model	Unique Terms	Exact Matches	Non-Matches	Similar Terms (Excl. Exact)
gemma3_12b	104,515	20,082	78,792	5,641
gemma3n_e4b	169,462	31,258	127,133	11,071
granite3_3_8b	164,864	19,467	136,678	8,719
phi4-reasoning_14b	151,213	22,179	118,101	10,933
qwen2_5_7b	145,225	20,930	113,010	11,285
qwen2_5_14b	137,611	20,471	106,810	10,330

Table 3: Comparison between LLM and spaCy NER Models — SOAP

Assessment

LLM Model	Unique Terms	Exact Matches	Non-Matches	Similar Terms (Excl. Exact)
gemma3_12b	26,174	8,359	13,323	4,492
gemma3n_e4b	27,553	7,804	14,762	4,987
granite3_3_8b	37,477	4,617	28,251	4,609
phi4-reasoning_14b	31,683	3,760	22,037	5,886
qwen2_5_7b	33,058	827	24,590	7,641
qwen2_5_14b	30,183	4,226	19,109	6,848

Table 4: Comparison between LLM and spaCy NER Models — SOAP Assessment Notes

LLM Model	Unique Terms	Exact Matches	Non-Matches	Similar Terms (Excl. Exact)
gemma3_12b	16,428	5,776	9,200	1,452
gemma3n_e4b	18,724	5,272	11,363	2,089
granite3_3_8b	18,281	4,369	11,862	2,050
phi4-reasoning_14b	19,156	3,102	12,656	3,398
qwen2_5_7b	18,539	2,828	11,902	3,809
qwen2_5_14b	17,946	3,622	11,207	3,117

Conclusion and future directions

- Development of comprehensive medication knowledge base from active pharmaceutical ingredients to products to brand drug names interlinked with national standards to global community standards and databases through OMOP CDM
- Development of computational workflows for automatic extraction of clinical terms (medication, symptom/signs, and diseases) from unstructured EHR notes using open source small language models and rule-based NLP algorithms
- Benchmarking of language models highlights fine-tuning models outperformed prompt-based methods
- Integration of Human-In-The-Loop (HITL) Intelligent Systems would be promising solutions to accelerate the automatic standardization and harmonization of large scale unstructured clinical notes

Contributors & Acknowledgements

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